Single-mode fidelity of collective photon emission

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Abstract

For ideal Dicke emission from N initially excited emitters, the maximum squared overlap with any single-temporal-mode N-photon Fock state is $\Theta(N^{-1/2})$. An explicit normalized pulse and a relative-entropy inequality give the lower bound; temporal photon counts give an upper bound uniform over all pulse shapes. The argument extends to n symmetric excitations in M emitters: near full inversion the fidelity is $\Theta(\sqrt{(M-n+1)/M})$, whereas it tends to one when $n/M \to 0$. Uniform attenuation retains a conditional ceiling of order $k^{-1/2}$ for k surviving photons, and at fixed transmission the best unconditional photon-number and mode target has yield $\Theta(N^{-1})$. Independent local decay instead produces a different exact conditional kernel and a polynomial all-photon yield at fixed loss ratio. Diagonal collective phases and common rate shaping cannot improve the ideal modal fidelity. Detecting and consuming a prefix gives an exact resource law, with explicit detection-efficiency and switching-delay costs. Exact Fock emission from a coherent Markov ladder requires bosonic rates and separable phases; under a native-rate cap exact purity is slow. This exactness restriction is singular: a capped, excitation-dependent coherent coupling achieves fidelity tending to one at the native order of completion time, with polynomial all-photon acceptance at fixed local loss. The control construction specifies a mathematical coupling model rather than an experimental implementation.

1 Model, observables, and main result

We consider identical two-level emitters in their fully symmetric spin sector, vacuum input, one desired Markov output, and complete eventual emission. The native coupling is $L_{\mathrm{out}} = \sqrt{\Gamma} J_-$, with $\Gamma > 0$. Unless a section explicitly introduces loss or control, there is no pumping, dephasing, feedback, or other output channel. These are the ideal collective-emission assumptions used in the waveguide descriptions of Refs. [1, 2].

Let $b^\dagger(t)$ create an output photon, with $[b(t), b^\dagger(s)] = \delta(t-s)$. For a normalized temporal mode,

$$a_f^\dagger = \int_0^\infty f(t) b^\dagger(t)\,\mathrm{d}t, \qquad |n_f\rangle = \frac{(a_f^\dagger)^n}{\sqrt{n!}}|0\rangle, \qquad F^\star(\rho_n) = \sup_{\|f\|_2=1} \langle n_f|\rho_n|n_f\rangle. \tag{1}$$

Here ρ_n has trace one in the n-photon sector. All fidelities are squared overlaps, or their mixed-state extension against a pure target. The optimization includes arbitrary complex pulse shapes and is not restricted to the most populated one-photon mode. If the sector occurs with probability p_n, its unconditional target-state yield is $Y_n^\star = p_n F^\star(\rho_n)$. Sector probability, conditional mode fidelity, and yield are separate quantities.

For the lossless native and partial-source calculations in Sections 2–6, time is measured in units Γ^{-1}, and t denotes the dimensionless coordinate. Rescaling a physical mode by $\widetilde{f}(\tau) = \Gamma^{-1/2} f(\tau/\Gamma)$ preserves its norm and all overlaps. Later loss and control sections use physical t, with dimensionless $\tau = \Gamma t$ stated when needed. Euler's gamma function is denoted Γ_E, to distinguish it from the emission rate.

Theorem 1 (Fully inverted source)**.** *For the ideal output* $\rho_N = |\Psi_N\rangle\langle\Psi_N|$ *of* N *fully excited emitters, write* $F_N^\star = F^\star(\rho_N)$*. Then*

$$\boxed{\sqrt{2\pi} \le \liminf_{N\to\infty} \sqrt{N} F_N^\star \le \limsup_{N\to\infty} \sqrt{N} F_N^\star \le \frac{4\sqrt{2\pi}}{e}.} \tag{2}$$

Consequently $F_N^\star = \Theta(N^{-1/2})$*. The two endpoint constants are approximately* 2.506628 *and* 3.688548*.*

Equation (2) determines the power, not the existence or value of a limiting prefactor. The displayed asymptotic constants are not finite-N upper and lower inequalities without their limiting qualifications.

2 Exact emitted wavefunction

Use the normalized symmetric-coordinate convention

$$|\Psi_N\rangle = \frac{1}{\sqrt{N!}} \int_{[0,\infty)^N} \psi_N(\boldsymbol{t}) \prod_{i=1}^{N} b^\dagger(t_i)\, d^N t, \qquad \int |\psi_N|^2\, d^N t = 1. \tag{3}$$

The successive waiting times between emissions are independent exponentials with rates

$$\lambda_j = j(N-j+1), \qquad j = 1, \dots, N. \tag{4}$$

For ordered times $0 = t_0 < t_1 < \cdots < t_N$,

$$\sum_{j=1}^{N} \lambda_j (t_j - t_{j-1}) = \sum_{j=1}^{N} (2j - N) t_j \tag{5}$$

$$= \sum_i t_i + \sum_{i<j} |t_i - t_j|. \tag{6}$$

Since $\prod_j \lambda_j = (N!)^2$, the standard cascade amplitude [2] becomes

$$\boxed{\psi_N(\boldsymbol{t}) = \sqrt{N!} \exp\left[-\frac{1}{2} \sum_i t_i - \frac{1}{2} \sum_{i<j} |t_i - t_j| \right].} \tag{7}$$

Indeed the ordered-domain density is $N!|\psi_N|^2 = (N!)^2 \exp[-\sum_j \lambda_j (t_j - t_{j-1})]$, which integrates to one over that domain. The symmetric density $P_N = |\psi_N|^2$ therefore integrates to one over the full orthant. Positivity permits replacing f by $|f|$ without decreasing the overlap.

3 Constructive lower bound

3.1 A classical affinity identity

For a probability density p on the half-line, put $f = \sqrt{p}$ and $Q_p(\boldsymbol{t}) = \prod_i p(t_i)$. The amplitude is exactly the classical Hellinger affinity,

$$\langle N_f | \Psi_N \rangle = \int \sqrt{P_N Q_p}\, d^N t = \mathbb{E}_{Q_p} \exp\left[\tfrac{1}{2} \log(P_N / Q_p) \right]. \tag{8}$$

Jensen's inequality, followed by squaring, gives

$$\boxed{F_N^\star \geq F_N[f] \geq \exp[-D(Q_p\|P_N)].} \tag{9}$$

From Eq. (7),

$$D(Q_p\|P_N) = N\int p\log p + N\mathbb{E}_p t + \frac{N(N-1)}{2}\mathbb{E}_p|t-s| - \log N!. \tag{10}$$

No approximation or independence assumption about the actual output is used: only the trial density Q_p is a product.

3.2 An explicit normalized pulse

Choose

$$\boxed{p_N(t) = \frac{N^2 e^{Nt}}{(e^{Nt}+N-1)^2}, \qquad f_N(t) = \frac{Ne^{Nt/2}}{e^{Nt}+N-1}, \qquad t \geq 0.} \tag{11}$$

Its cumulative distribution is $(e^{Nt}-1)/(e^{Nt}+N-1)$, so normalization is exact. For $N > 1$,

$$\mathbb{E}t = \frac{\log N}{N-1}, \qquad \mathbb{E}|t-s| = \frac{2[(N-1)-\log N]}{(N-1)^2}, \tag{12}$$

$$\int p_N \log p_N = \frac{N\log N}{N-1} - 2. \tag{13}$$

Substitution yields

$$D(Q_{p_N}\|P_N) = \frac{N^2\log N}{N-1} - N - \log N!, \tag{14}$$

and the exact finite-N lower bound

$$\boxed{F_N^\star \geq L_N := \frac{e^N N!}{N^{N^2/(N-1)}} \quad (N \geq 2).} \tag{15}$$

For $N = 1$, Eq. (11) reduces to $p_1(t) = e^{-t}$ and $F_1^\star = 1$. Stirling's expansion gives

$$L_N = \sqrt{\frac{2\pi}{N}}\exp\left[-\frac{\log N}{N-1} + \frac{1}{12N} + O(N^{-3})\right], \tag{16}$$

which proves the lower half of Eq. (2).

3.3 Optimality within the relative-entropy variational principle

The pulse in Eq. (11) globally minimizes Eq. (10), although it need not maximize the true fidelity. Let $X(u)$ be the quantile of p, $v(u) = X'(u) > 0$, and $x_0 = X(0) \geq 0$. For an absolutely continuous quantile with finite objective,

$$\frac{D(Q_p\|P_N) + \log N!}{N} = x_0 + \int_0^1 [A_N(u)v(u) - \log v(u)]\,du, \tag{17}$$

where $A_N(u) = (1-u)[1+(N-1)u]$. The integrand is strictly convex in v and minimized pointwise at $v = 1/A_N$; x_0 is minimized at zero. Integration gives

$$X_N(u) = \frac{1}{N}\log\frac{1+(N-1)u}{1-u}, \tag{18}$$

equivalent to Eq. (11). Quantile jumps and singular increments add the nonnegative Stieltjes term $\int A_N\,dX_{\text{singular}}$ and cannot improve the infimum. This establishes the same minimum for density-derived generalized quantiles with finite objective.

4 Uniform upper bound from temporal photon counts

4.1 Every trial pulse supplies its own median cut

Fix any normalized mode f. There is a time T with $\int_0^T |f(t)|^2 dt = 1/2$. The number m of photons before T in $|N_f\rangle$ has distribution

$$b_N(m) = 2^{-N}\binom{N}{m}. \tag{19}$$

Let $p_m(T)$ be the corresponding Dicke-output count distribution. Projection onto the orthogonal count sectors, followed by Cauchy–Schwarz within each sector, implies

$$|\langle N_f|\Psi_N\rangle| \le \sum_{m=0}^{N} \sqrt{p_m(T) b_N(m)}. \tag{20}$$

The cut may depend on f because the following estimates are uniform in T. This is an analysis device, not an assumption of feedback during emission.

4.2 Arrival times bound count probabilities

The emitted count is a pure-birth process with rates $r_m = (m+1)(N-m)$ from m to $m+1$. This reduction is standard Dicke dynamics [4]. Write T_m for the mth arrival and h_m for its density. For $1 \le m \le N-1$,

$$p_m(T) = \int_0^T h_m(s) e^{-r_m(T-s)}\, ds \le \frac{\|h_m\|_\infty}{r_m}. \tag{21}$$

T_m contains the independent first waiting time $\mathrm{Exp}(N)$, so convolution gives $\|h_m\|_\infty \le N$. Already,

$$p_m(T) \le \min\left\{1, \frac{N}{(m+1)(N-m)}\right\}. \tag{22}$$

Together with Eq. (20), this gives an explicit finite-N upper bound by inserting the displayed expression for $1 \le m < N$ and the bound one for $m = 0, N$, and clipping the resulting squared sum at one.

To improve the asymptotic constant, fix $K \ge 2$ independently of N. For $m \ge K$, convolution contraction gives $\|h_m\|_\infty \le \|h_K\|_\infty$. The density of NT_K converges uniformly to that of $Y_K = \sum_{j=1}^K \mathrm{Exp}(j)$. This follows from convergence of each fixed-rate exponential density in both L^1 and L^∞, followed by convolution. Y_K has the distribution of the maximum of K independent unit-rate exponentials, hence

$$h_{Y_K}(s) = Ke^{-s}(1-e^{-s})^{K-1}, \qquad \|h_{Y_K}\|_\infty = c_K = (1-1/K)^{K-1}. \tag{23}$$

Therefore $\|h_K\|_\infty = N[c_K + o_N(1)]$. Uniformly over T and $|m - N/2| \le N^{2/3}$,

$$p_m(T) \le \frac{4c_K + o_N(1)}{N}. \tag{24}$$

4.3 The binomial width determines the exponent

A binomial Chernoff bound and Cauchy–Schwarz give

$$\sum_{|m-N/2|>N^{2/3}} \sqrt{p_m(T) b_N(m)} \le \sqrt{2}\, e^{-N^{1/3}}. \tag{25}$$

Stirling's formula with Gaussian tail control gives

$$\sum_{m=0}^{N} \sqrt{b_N(m)} = (2\pi N)^{1/4}[1 + o(1)]. \tag{26}$$

Combining the central and tail estimates in Eq. (20), uniformly over every f,

$$F_N^\star \le \frac{4c_K\sqrt{2\pi} + o_N(1)}{\sqrt{N}}. \tag{27}$$

First take $N \to \infty$ at each fixed K, then $K \to \infty$. Since $c_K \to e^{-1}$, the upper half of Eq. (2) follows. No exchange of a growing-K limit is needed.

4.4 Physical origin of the count bound

The source starts at rate N and accelerates to rates of order N^2. The early waiting-time uncertainty is not erased by that acceleration. At a trial mode's median cut, a Fock pulse has a binomial count width of order $\sqrt{N}$; Dicke emission has central count probabilities of order $1/N$. This mismatch bounds the affinity even after pulse optimization. The arrival-density divided by dwell-rate argument also applies to other radiative cascades, although it alone does not supply a matching lower exponent for another source.

5 Dependence on the initial excitation

Let M be the total number of emitters and n the excitation number of an initially symmetric Dicke state. Put $a = M - n + 1$. Exciting an arbitrary subset of n emitters in a larger array does not in general prepare this state or preserve the symmetric sector. In the lossless symmetric problem the emitted-count rates are $r_j = (n - j)(a + j)$, $j = 0, \dots, n - 1$. Their product is $n!\Gamma_E(M + 1)/\Gamma_E(a)$, so the same ordered-coordinate normalization as in Section 2 gives

$$P_{M,n}(\boldsymbol{t}) = \frac{\Gamma_E(M+1)}{\Gamma_E(a)} \exp\left[-a\sum_i t_i - \sum_{i<j} |t_i - t_j|\right]. \tag{28}$$

Its positive square root is the normalized emitted amplitude.

5.1 An exact product relative-entropy bound

The quantile argument of Section 3 now has $A(u) = (1 - u)[a + (n - 1)u]$, and its initial-offset term is ax_0. The pointwise minimizing derivative is again $X'(u) = 1/A(u)$. For $n > 1$,

$$X_{M,n}(u) = \frac{1}{M} \ln \frac{a + (n-1)u}{a(1-u)}, \tag{29}$$

$$p_{M,n}(t) = \frac{aM^2 e^{Mt}}{[ae^{Mt} + n - 1]^2}. \tag{30}$$

The cumulative distribution of this pulse is $a(e^{Mt} - 1)/(ae^{Mt} + n - 1)$. Thus normalization is exact. Its product relative entropy with respect to (28) is

$$\boxed{D_{M,n} = n\left[\frac{M\ln M - a\ln a}{n-1} - 1\right] - \ln\Gamma_E(M+1) + \ln\Gamma_E(a), \qquad F_{M,n}^\star \ge e^{-D_{M,n}}.} \tag{31}$$

The formula follows by substituting the quantile into the same strictly convex functional used for the fully inverted source. It is the global minimum of the product relative entropy, not necessarily the pulse maximizing the actual fidelity. For $n = 1$, the density is $p(t) = Me^{-Mt}$, and fidelity one is exact.

Equation (28), its normalization, and the variational calculation also hold for integer n and any real $a \geq 1$, with $M = n + a - 1$. In that extension M is a distribution parameter, not a fractional emitter number. This will be needed for local-loss conditioning in Section 8.

5.2 Uniform control in the dilute regime

Set $b = n - 1$. Equation (31) is equivalently

$$D_{M,n} = n \left[\frac{1}{b} \int_a^M \ln x \, \mathrm{d}x - \frac{1}{b+1} \sum_{j=0}^{b} \ln(a+j) \right]. \tag{32}$$

The equally weighted, endpoint-inclusive grid dominates the uniform interval in convex order. To see this, a convex function's integral average is at most its composite trapezoidal average. If its grid values are $g_0, \ldots, g_b$, the grid average minus the trapezoidal average is

$$\frac{(b-1)(g_0+g_b)/2 - \sum_{j=1}^{b-1} g_j}{b(b+1)} \geq 0, \tag{33}$$

where the inequality follows by bounding the internal values by the endpoint chord. For $x^2/2$, the grid average minus the continuous average is $b/12$. Comparing the curvature of $\ln x$ with $-M^{-2}$ and $-a^{-2}$, or applying convex order to $\ln x + x^2/(2a^2)$ and concave order to $\ln x + x^2/(2M^2)$, therefore gives

$$\boxed{\frac{n(n-1)}{12M^2} \leq D_{M,n} \leq \frac{n(n-1)}{12a^2}.} \tag{34}$$

Consequently,

$$F^\star_{M,n} \geq \exp\left[-\frac{n(n-1)}{12(M-n+1)^2} \right], \qquad n/M \to 0 \implies F^\star_{M,n} \to 1. \tag{35}$$

This remains uniform when $n \to \infty$. The qualitative weak-excitation route to Fock emission is established in Ref. [6]; (35) quantifies the present optimized-pulse problem.

5.3 Fixed excitation fraction and uniform near-inversion scaling

Write the Stirling remainder as

$$\ln \Gamma_E(x) = (x - \tfrac{1}{2}) \ln x - x + \tfrac{1}{2} \ln(2\pi) + R(x), \qquad 0 < R(x) < \frac{1}{12x}, \quad x \geq 1. \tag{36}$$

Using $\Gamma_E(M+1) = M\Gamma_E(M)$, Eq. (31) rearranges exactly to

$$D_{M,n} = \frac{1}{2} \ln \frac{M}{a} - 1 + \frac{a}{n-1} \ln \frac{M}{a} + R(a) - R(M). \tag{37}$$

In particular, $\ln(1+z)/z \leq 1$ gives the real-a bound

$$F^\star_{M,n} \geq e^{-D_{M,n}} \geq e^{-1/(12a)} \sqrt{\frac{a}{M}}. \tag{38}$$

If $n/M \to \rho \in (0,1)$, Eq. (37) gives

$$\liminf F^\star_{M,n} \geq e(1-\rho)^{(2-\rho)/(2\rho)} > 0. \tag{39}$$

This is a lower bound, not the exact limiting fidelity. At $\rho = 1/2$ it is $e/(2\sqrt{2}) = 0.961057757\ldots$.

Theorem 2 (Near complete inversion, including real a)**.** *For integer n, real $a \geq 1$, and $M = n + a - 1$, along every sequence $M \to \infty$ with $a = o(M)$,*

$$\boxed{F^{\star}_{M,n} = \Theta\left(\sqrt{\frac{a}{M}}\right).} \tag{40}$$

The implicit constants may be chosen uniformly along these sequences. For physical integer M, n, this is the symmetric partial-source fidelity.

Proof. The lower estimate is uniform by (38). For the upper estimate take $J = \max(2, \lceil a \rceil)$, which is $o(n)$. Each of the first J waiting rates is at most $2na$. The sum T_J of those waits therefore has characteristic function bounded by

$$|\widehat{h}_J(\omega)| \leq [1 + (\omega/(2na))^2]^{-J/2}. \tag{41}$$

Fourier inversion gives

$$\|h_J\|_\infty \leq \frac{na}{\sqrt{\pi}} \frac{\Gamma_E((J-1)/2)}{\Gamma_E(J/2)} \leq \frac{na}{\sqrt{J-1}} \leq \sqrt{2}\, n\sqrt{a}. \tag{42}$$

The middle inequality is the beta-integral bound $\int_{\mathbb{R}} (1+x^2)^{-J/2}\, \mathrm{d}x \leq \pi/\sqrt{J-1}$; it follows from the Wallis gamma-ratio inequality and is equality at $J = 2$. Convolution with later waits does not increase the density supremum. The dwell bound (21) now yields

$$p_m(T) \leq \frac{\sqrt{2}\, n\sqrt{a}}{(n-m)(a+m)}, \qquad m \geq J. \tag{43}$$

For $n/4 \leq m \leq 3n/4$ this is $O(\sqrt{a}/n)$. At the median of an arbitrary trial pulse, the target count is $\mathrm{Bin}(n, 1/2)$. Its square-root probabilities sum to $O(n^{1/4})$; the contribution outside the stated central interval is exponentially small by Cauchy–Schwarz and the binomial tail bound. Equation (20) therefore gives $F^{\star}_{M,n} = O(\sqrt{a/n})$. Since $n/M \to 1$, this proves (40). □

The trial lower bound has more explicit constants. If $a \to \infty$ while $a = o(M)$, (37) gives $e^{-D_{M,n}} = [e + o(1)]\sqrt{a/M}$. If a is fixed,

$$\sqrt{M}\, e^{-D_{M,n}} \longrightarrow \sqrt{2\pi} \frac{a^a e^{1-a}}{\Gamma_E(a)}. \tag{44}$$

These are asymptotics of the analytic lower bound, not asserted optimal-fidelity prefactors. They include the fully inverted case $a = 1$. At the opposite extreme $n/a \to 0$, (38) also implies fidelity tending to one.

6 Exact optimal fidelity for two photons

For $N = 2$, $\psi_2(t, s) = \sqrt{2} e^{-\max(t,s)}$ is a positive symmetric integral kernel. Its eigenvalue equation differentiates to

$$f'' + f' + \frac{\sqrt{2}}{\lambda} e^{-t} f = 0, \qquad f'(0) = 0. \tag{45}$$

Decay at infinity selects the Bessel solution. If $j_{0,1}$ is the first positive zero of J_0,

$$\boxed{F^{\star}_2 = \frac{32}{j^4_{0,1}} = 0.956787875131927\ldots, \qquad f^{\star}_2(t) = \frac{e^{-t/2} J_1(j_{0,1} e^{-t/2})}{J_1(j_{0,1})}.} \tag{46}$$

The pulse is exactly normalized. The first zero gives the largest kernel eigenvalue $4\sqrt{2}/j^2_{0,1}$.

7 Uniform attenuation: surviving photons retain a modal ceiling

Let $\mathcal{L}_\eta$ be uniform, mode-independent attenuation of the ideal output, with vacuum in the unused port and $0 < \eta < 1$. The number K_{out} of transmitted photons is binomial:

$$\mathbb{P}(K_{\text{out}} = k) = B_{N,\eta}(k) = \binom{N}{k}\eta^k(1-\eta)^{N-k}. \tag{47}$$

Conditioned on k, the temporal state is the normalized k-particle reduced density operator $\rho_N^{(k)}$. This follows by applying the same beam splitter to each particle coordinate: choosing the transmitted labels gives the binomial coefficient, and tracing the other port traces the remaining coordinates. Write $F^\star_{N\to k} = F^\star(\rho_N^{(k)})$.

Theorem 3 (Uniform survivor bound)**.** *There is a universal $C < \infty$ such that $F^\star_{N\to k} \le C/\sqrt{k}$ for all $N \ge k \ge 1$. More precisely, along every sequence $N \ge k \to \infty$,*

$$\limsup \sqrt{k}\, F^\star_{N\to k} \le \frac{4\sqrt{2\pi}}{e}. \tag{48}$$

For a fixed retained fraction $k/N \to \eta_0 \in (0,1]$, $F^\star_{N\to k} = \Theta(N^{-1/2})$.

Proof. Let $p_m(T)$ be the probability of m original photons before a dimensionless cut T. Section 4 already establishes the arrival-density and dwell estimate

$$p_m(T) \le \frac{N}{(m+1)(N-m)} \le \frac{N}{m(N-m)}, \qquad 1 \le m < N. \tag{49}$$

The retained early count has the hypergeometric mixture

$$q_j(T) = \sum_{m=0}^{N} p_m(T) \underbrace{\frac{\binom{m}{j}\binom{N-m}{k-j}}{\binom{N}{k}}}_{\mathrm{HG}(N,m,k,j)}. \tag{50}$$

For $1 \le j < k$, the kernel vanishes at $m = 0, N$. The identities $\binom{m}{j}/m = \binom{m-1}{j-1}/j$ and its complementary version, followed by Vandermonde convolution, give

$$\sum_{m=1}^{N-1} \frac{\mathrm{HG}(N,m,k,j)}{m(N-m)} = \frac{\binom{N-1}{k-1}}{j(k-j)\binom{N}{k}} = \frac{k}{Nj(k-j)}, \tag{51}$$

$$q_j(T) \le \frac{k}{j(k-j)}. \tag{52}$$

This estimate is uniform in T, N, k.

For any proposed mode choose a median time T of $|f|^2$. The target $|k_f\rangle$ has cut-count law $b_j = \binom{k}{j}2^{-k}$. Monotonicity of fidelity under this measurement implies

$$\langle k_f|\rho_N^{(k)}|k_f\rangle \le \left(\sum_{j=0}^{k}\sqrt{q_j(T)b_j}\right)^2. \tag{53}$$

On $|j - k/2| \le k^{2/3}$, Eq. (52) is at most $(4 + o(1))/k$. Stirling's formula gives $\sum_j \sqrt{b_j} \sim (2\pi k)^{1/4}$. Outside this window, the affinity contribution is at most the square root of the binomial tail probability, by Cauchy–Schwarz, and is exponentially small in $k^{1/3}$. This proves a universal $O(k^{-1/2})$ bound.

For the stated prefactor, Section 4 proves that the density supremum of the first fixed $J \geq 2$ arrivals is $N[c_J + o_N(1)]$, where $c_J = (1-1/J)^{J-1}$. Thus the numerator N in (49) can be replaced by $N[c_J + o_N(1)]$ whenever $m \geq J$. For central retained $j \geq J$, every contributing m is at least J, so the same Vandermonde identity applies. Equation (53) yields $\limsup \sqrt{k} F^{\star}_{N\to k} \leq 4c_J\sqrt{2\pi}$. Since $N \geq k \to \infty$, the error is uniform along every stated sequence. Taking $J \to \infty$ gives (48).

For the lower bound, the falling-factorial occupation identity is

$$\langle k_f|\rho_N^{(k)}|k_f\rangle = \left\langle \frac{(\hat{n}_f)_k}{(N)_k} \right\rangle \geq \langle N_f|\rho_N|N_f\rangle. \tag{54}$$

The contribution from occupation $n_f = N$ is one and all other terms are nonnegative. Optimize and apply (2). For k proportional to N, the two orders match. □

Corollary 4 (Optimize both output photon number and mode)**.** *At fixed* $0 < \eta < 1$*,*

$$\max_{0\leq k\leq N} \sup_f \langle k_f|\mathcal{L}_\eta(\rho_N)|k_f\rangle = \Theta(N^{-1}). \tag{55}$$

If all N photons are required instead, the yield is $\eta^N F^{\star}_N = \Theta(\eta^N/\sqrt{N})$.

Proof. In a fixed bulk interval around ηN, the binomial point probability is $O(N^{-1/2})$ and the conditional fidelity is $O(N^{-1/2})$. Outside that interval the point probability is exponentially small. For the reverse inequality choose a modal binomial count, whose probability is $\Theta(N^{-1/2})$, and use (54) with the full-source trial mode. □

A parasitic *collective* channel $\sqrt{\kappa} J_-$ is equivalent to this beam-splitter loss after a clock rescaling, with $\eta = \Gamma/(\Gamma+\kappa)$. A time- or frequency-selective filter is not uniform attenuation and is not covered by this theorem.

8 Independent local decay and the cost of strong conditioning

Now add independent channels $L_i = \sqrt{\gamma}\,\sigma_i^-$, one for each emitter, with constant $\gamma \geq 0$, and define $\epsilon = \gamma/\Gamma$. With no pumping, observing all N desired photons means that no local jump occurred. The N-photon sector is pure: its local environment is vacuum and the emitters end in their ground state. It can be identified retrospectively by destructive total counting. A usable herald that leaves the photons available requires an additional nondestructive success flag or suitable monitoring of the undesired environment.

Proposition 5 (Exact all-photon sector)**.** *Put $a = 1+\epsilon$ and $M = N + a - 1$. The sector probability is*

$$p_0 = \prod_{j=1}^{N} \frac{j}{j+\epsilon} = \frac{\Gamma_E(N+1)\Gamma_E(1+\epsilon)}{\Gamma_E(N+1+\epsilon)}. \tag{56}$$

Its unnormalized symmetric amplitude and normalized density are

$$\psi_{\mathrm{un}}(\boldsymbol{\tau}) = \sqrt{N!}\, e^{-a\sum_i \tau_i/2 - \sum_{i<j}|\tau_i-\tau_j|/2}, \tag{57}$$

$$P_{\mathrm{cond}}(\boldsymbol{\tau}) = \frac{\Gamma_E(N+a)}{\Gamma_E(a)} e^{-a\sum_i \tau_i - \sum_{i<j}|\tau_i-\tau_j|}. \tag{58}$$

The physical-time amplitude additionally has the Jacobian factor $\Gamma^{N/2}$.

Proof. Before desired jump j, there are $q = N - j + 1$ excitations. The desired rate is Γjq, and the sum of the local rates is γq. The next jump is desired with probability $j/(j+\epsilon)$; its waiting time, independent of that jump identity, has rate $\Gamma q(j+\epsilon)$. Multiplying these probabilities proves (56). Along a history with no local jump, the integrated excitation number is $\sum_i t_i$. The no-local-jump propagator consequently multiplies the ideal amplitude by $e^{-\gamma \sum_i t_i/2}$, proving the first line. Dividing its squared modulus by (56) gives the second line. Equivalently the normalized ordered waits have dimensionless rates $(N-j+1)(a+j-1)$. □

The competing-rate, no-jump-product treatment of local decay is used by Paulisch *et al.* [1]. Its exact gamma quotient supplies the success probability and normalization for the optimized temporal-mode fidelity analyzed here.

Corollary 6 (Conditional scaling and fixed-loss target yield)**.** *For every sequence $a = 1+\epsilon \geq 1$ with $a = o(N)$,*

$$F^{\star}_{\mathrm{cond}} = \Theta\left(\sqrt{\frac{a}{N}}\right), \tag{59}$$

with uniform implicit constants. At fixed $\epsilon \geq 0$,

$$p_0 \sim \Gamma_E(1+\epsilon)N^{-\epsilon}, \qquad Y^{\star}_N = p_0 F^{\star}_{\mathrm{cond}} = \Theta(N^{-\epsilon-1/2}). \tag{60}$$

Proof. The normalized kernel (58) is precisely (28) with $n = N$, $a = 1+\epsilon$, and $M = N + a - 1$. The real-a theorem of Section 5 therefore gives the conditional scaling, with $M/N \to 1$ in this regime. The fixed-ϵ gamma-ratio asymptotic in (56) gives the sector probability, and multiplying proves the target-yield law. □

The unconditional optimized yield is nonincreasing in ϵ. Indeed an optimal mode for the nonnegative kernel can be chosen nonnegative, and increasing ϵ decreases every such unnormalized overlap pointwise. No corresponding monotonicity of the normalized conditional fidelity is assumed.

Corollary 7 (Strong conditioning has an exponential cost)**.** *If $\epsilon = cN$, with fixed $c > 0$, then*

$$\liminf F^{\star}_{\mathrm{cond}} \geq e\left(\frac{c}{1+c}\right)^{c+1/2}, \tag{61}$$

$$p_0 \sim \sqrt{\frac{2\pi N c}{1+c}}\left[\frac{c^c}{(1+c)^{1+c}}\right]^N. \tag{62}$$

More generally, keeping this native source's conditional fidelity bounded below by a positive constant requires $\epsilon = \Omega(N)$, and hence exponentially small acceptance. If $\epsilon/N \to \infty$, the conditional fidelity tends to one.

Proof. For $\epsilon = cN$, Eq. (37) tends to $(c+\frac{1}{2})\ln[(1+c)/c] - 1$; exponentiating proves (61). Stirling's formula in (56) gives (62). If ϵ were not $\Omega(N)$, a subsequence would have $a/N \to 0$, contradicting (40). When $\epsilon \geq cN$, monotonicity of p_0 gives an exponential upper bound using (62). Finally $a/N \to \infty$ makes the lower bound (38) tend to one. □

At $\epsilon = N$, these statements read $p_0 = \binom{2N}{N}^{-1} \sim \sqrt{\pi N}\, 4^{-N}$ and $\liminf F^{\star}_{\mathrm{cond}} \geq e/(2\sqrt{2}) = 0.961057757\ldots$. For size-dependent ϵ, the exact gamma quotient must be used outside the stated asymptotic regimes.

Local atomic loss is not uniform optical attenuation. After a local jump the emitter state can leave the symmetric sector, and the environment carries atom-specific information. For example, at $N = 2$ the desired photon-number probabilities are

$$\mathbb{P}(K_{\text{out}} = 2) = \frac{2}{(1+\epsilon)(2+\epsilon)}, \quad \mathbb{P}(K_{\text{out}} = 1) = \frac{2\epsilon}{(1+\epsilon)(2+\epsilon)}, \quad \mathbb{P}(K_{\text{out}} = 0) = \frac{\epsilon}{2+\epsilon}. \tag{63}$$

To obtain them, after a first local jump decompose the remaining one-excitation mixture equally into its symmetric bright and antisymmetric dark states; only the bright part can emit into J_-. At $\epsilon = 1$ all three probabilities are $1/3$, which is not a two-photon binomial law. The survivor theorem of Section 7 therefore does not apply to the fewer-than-N sectors of local atomic loss.

The gamma-quotient formulas also extend to $\gamma(t) = \epsilon\Gamma(t)$ with constant ratio and complete emission, by using $\tau = \int_0^t \Gamma(s)\,\mathrm{d}s$. If $\Gamma(t)$ is shaped while physical γ stays fixed, the factor $e^{-\gamma\sum_i t_i/2}$ remains in physical time; the constant-ratio formulas cannot be substituted without a new calculation.

9 Controls that cannot improve the modal fidelity

Label the symmetric ladder by the emitted count $m = 0, \ldots, N$. Consider a single coherent output and a diagonal Hamiltonian,

$$L(t) = \sum_{m=0}^{N-1} \sqrt{r_m(t)}\, e^{i\theta_m(t)} |m+1\rangle\langle m|, \qquad H(t) = \sum_{m=0}^{N} E_m(t) |m\rangle\langle m|. \tag{64}$$

We set $\hbar = 1$. Every Hamiltonian that preserves both excitation number and the fully symmetric spin sector has this diagonal form, because each excitation sector of that spin representation is one-dimensional. This statement does not cover Hamiltonians that mix spin sectors or introduce a cavity, memory, or other ancilla.

Proposition 8 (Phase removal and scalar rate shaping)**.** *At fixed rates $r_m(t)$, removing the Hamiltonian and jump phases cannot decrease the optimized full-photon fidelity or the optimized fidelity of any normalized k-particle reduced state. For native rates $r_m(t) = \kappa(t)(m+1)(N-m)$, common deterministic shaping with $\int_0^\infty \kappa(t)\,\mathrm{d}t = \infty$ leaves those optimized fidelities unchanged.*

Proof. Let $\psi_0 \geq 0$ be the phase-free amplitude. Equation (64) changes it only to $\psi_H(\boldsymbol{t}) = \psi_0(\boldsymbol{t}) e^{i\Phi(\boldsymbol{t})}$. For retained variables $\boldsymbol{t}$ and traced variables $\boldsymbol{z}$,

$$F_k(H; f) = \int \mathrm{d}\boldsymbol{z} \left| \int \psi_H(\boldsymbol{t}, \boldsymbol{z}) \prod_{i=1}^{k} f(t_i)^* \,\mathrm{d}\boldsymbol{t} \right|^2 \tag{65}$$

$$\leq \int \mathrm{d}\boldsymbol{z} \left[\int \psi_0(\boldsymbol{t}, \boldsymbol{z}) \prod_{i=1}^{k} |f(t_i)|\,\mathrm{d}\boldsymbol{t} \right]^2 = F_k(0; |f|). \tag{66}$$

For $k = N$ there are no traced variables. Optimization proves the phase claim, including after uniform attenuation.

Let $\tau(t) = \int_0^t \kappa(s)\,\mathrm{d}s$. The map

$$(U_\kappa f)(t) = \sqrt{\kappa(t)}\, f(\tau(t)) \tag{67}$$

is a one-photon isometry, and acts identically on every emitted coordinate. The transformed field and all its reduced states lie in its range, so optimization over all physical modes gives the same fidelity as before the transformation. Partial tracing commutes with this common isometry. This proves the rate claim. □

Arbitrary functions of J_z, including nonlinear collective phases within this ladder, are covered by (66). Combining them with common rate shaping cannot remove (2) or (48). The same invariance holds for any common lossless one-photon unitary. Independent local loss at a fixed physical rate changes the problem, as the last paragraph of Section 8 explains.

9.1 Ideal all-photon mode filtering

For an ideal filter retaining one normalized temporal mode f and rejecting its orthogonal complement, the probability that all N photons are retained is $|\langle N_f|\Psi_N\rangle|^2$. On that event the retained state is $|N_f\rangle$. Optimization gives $F_N^\star = \Theta(N^{-1/2})$ for the fully inverted source. This equality follows by projecting onto occupation $n_f = N$. It gives an all-photon extraction probability, distinct from a deterministic source guarantee. A usable herald must identify success without consuming the desired retained photons; the projection probability alone does not supply such an apparatus.

10 Detect a prefix: an exact resource and imperfection law

Start with the lossless native source. Detect and consume its first r photons, $0 \le r < N$, and immediately route all later photons to the retained output. Use the rth-click time as the new time origin; for $r = 0$, use the preparation time.

Proposition 9 (Prefix identity)**.** *With perfect detection, time resolution, and switching, the record-aligned retained field has exactly the partial-source state with $M = N$, $n = N-r$, $a = r+1$. Hence*

$$F_{\mathrm{prefix}}(N,r) = F^\star_{N,N-r}. \tag{68}$$

In particular, for $r = o(N)$,

$$F_{\mathrm{prefix}}(N,r) = \Theta\left(\sqrt{\frac{r+1}{N}}\right). \tag{69}$$

If $r/N \to \alpha \in (0,1)$, then

$$\liminf F_{\mathrm{prefix}}(N,r) \ge e\,\alpha^{(1+\alpha)/[2(1-\alpha)]}. \tag{70}$$

Proof. Between clicks the symmetric ladder state acquires a scalar no-jump factor; each click applies J_-. After r desired jumps the normalized state is the symmetric Dicke state with $N-r$ excitations, independent of all earlier waiting times. The Markov property and stationary coupling make the future field relative to that click precisely the partial-source output. The remaining assertions follow from Section 5. No favorable timing record is selected: in the ideal model the rth click eventually occurs with probability one. □

A sublinear consumed prefix cannot keep the fidelity bounded away from zero. A macroscopic prefix permits a nonzero limiting fidelity while retaining a macroscopic photon number. For $\alpha = 1/2$, the explicit lower bound is $e/(2\sqrt{2}) = 0.961057757\ldots$, not the exact optimal fidelity. If $N - r = o(N)$, the dilute-source bound implies fidelity tending to one. At finite size with $N - r > 1$, one may use $e^{-D_{N,N-r}}$. For one retained photon the fidelity is exactly one. A simpler sufficient bound, valid also in that case, is

$$F_{\mathrm{prefix}}(N,r) \ge \exp\left[-\frac{(N-r)(N-r-1)}{12(r+1)^2}\right]. \tag{71}$$

Ignoring the recorded origin produces a mixture of temporal translations, a different target problem. This protocol consumes the prefix; it is not a nondestructive measurement on the original N-photon state, nor an optimality theorem for all adaptive controls.

Proposition 10 (Exact nominal-target detection and latency cost)**.** *Let $r \geq 1$, detector efficiency $0 < d \leq 1$, and a deterministic delay $\Delta \geq 0$ between the rth detected click and routing. Each prefix photon is independently detected, no other source loss is present, and native emission continues during the delay. For the predetermined target photon number $n = N - r$, the optimized record-aligned yield per initial trial is exactly*

$$Y_{N-r}^{\mathrm{nom}}(d, \Delta) = d^r e^{-\Gamma(N-r)(r+1)\Delta} F^{\star}_{N,N-r}. \tag{72}$$

Conditioning the ensemble on a trigger divides this by $\mathbb{P}[\mathrm{Bin}(N, d) \geq r]$.

Proof. If S photons have actually been emitted at the rth detected click, then

$$\mathbb{P}(S = s, \mathrm{trigger}) = \binom{s-1}{r-1} d^r (1-d)^{s-r}, \qquad r \leq s \leq N. \tag{73}$$

Summing gives the stated trigger probability. Only $S = r$ can leave the nominal $N - r$ photons, and that event has probability d^r. From this post-click level, the next waiting rate is $\Gamma(N - r)(r + 1)$. Keeping the same nominal number during the delay requires no additional emission, with probability equal to the exponential factor in (72). Conditioned on no such jump, the normalized spin state is unchanged. Its future field, referred to the switch time, is the same partial source. Every other branch is orthogonal to the nominal photon-number target, so multiplication gives the exact result. □

For $r = \Theta(N)$, fixed $d < 1$ gives an exponential nominal-yield penalty. To keep the detection factor nonvanishing requires $r[-\ln d] = O(1)$, equivalently $r(1 - d) = O(1)$ near unit efficiency. When both r and $N - r$ are proportional to N, the delay factor requires $\Gamma N^2 \Delta = O(1)$; in general the exact condition is $\Gamma(N - r)(r + 1)\Delta = O(1)$. The corresponding $o(1)$ conditions make the penalties tend to one. These statements apply to this stopping rule and predetermined target. Extra number information, accepting several output counts, buffering, or stopping emission during the delay changes the protocol and its resource accounting.

11 Exact single-mode ladders and an exact-purity time penalty

We return to the coherent ladder (64), with locally integrable rates, vacuum input, and complete emission. No extra environment records its excitation-dependent control. For a normalized target pulse $p(t) = |f(t)|^2$, define its survival $S(t) = 1 - \int_0^t p(s)\,\mathrm{d}s$ and hazard $h(t) = p(t)/S(t)$ where $S(t) > 0$.

Theorem 11 (Exact Fock criterion within a Markov ladder)**.** *The output is exactly $|N_f\rangle$, up to a global phase, if and only if on reachable times and levels*

$$r_m(t) = (N - m)h(t), \qquad \phi_m(t) = \arg f(t) + c_m \pmod{2\pi}, \tag{74}$$

where c_m are constants and

$$\phi_m(t) = \theta_m(t) - \int_0^t [E_m(s) - E_{m+1}(s)]\,\mathrm{d}s. \tag{75}$$

The rate equality holds almost everywhere at every reachable time, including pulse gaps where $h = 0$. The phase equality is required only where $f \neq 0$. When its cumulative probability is strictly between zero and one, every level is reachable and the rate equality applies to every $m = 0, \ldots, N - 1$.

Proof. In an N-photon Fock mode the detection times are independent draws from p. Given m arrivals by time t, the hazard of the next is $(N-m)h(t)$. The physical ladder hazard is $r_m(t)$. Equality of the joint density therefore forces the first condition. All levels have positive binomial probability when the cumulative probability lies in $(0,1)$. Conversely, those hazards generate the independent order-statistic law, so the positive amplitude equals the target amplitude.

For ordered times $0 < t_1 < \cdots < t_N$, the product of jump phases and diagonal propagators has phase $\sum_{j=1}^{N} \phi_{j-1}(t_j)$, up to a global final-state phase. To equal the target, its difference from $\sum_j \arg f(t_j)$ must be constant on the ordered support. Varying one ordered coordinate at a time implies the second condition almost everywhere; the constants are consistent across overlapping ordered neighborhoods. The condition is sufficient by multiplication. Support gaps and times after release impose only the corresponding reachable-state requirements. □

For stationary positive rates and phases removed, the criterion reduces to $r_m = \kappa(N-m)$, with $f(t) = \sqrt{\kappa} e^{-\kappa t/2}$. A formal realization is

$$L_{\mathrm{lin}}(t) = \sqrt{h(t)}\, J_-[N - \hat{n} + 1]^{-1/2}, \tag{76}$$

where $\hat{n}$ is the current excitation number and the right factor acts first. Its matrix element is $\sqrt{h(t)n}$. This is a truncated bosonic annihilation ladder. Bosonic Fock-state release and pulse synthesis are established prior constructions [10, 11, 12]; Eq. (76) identifies the coupling required inside this spin ladder. Measuring the count and updating a scalar rate is not automatically equivalent: an extra history record can destroy temporal coherence.

For reference, a stationary positive ladder has the following exact overlap with the exponential comparison mode:

$$F_{\exp}(\kappa) = \prod_{m=0}^{N-1} \left[\frac{2\sqrt{r_m \kappa (N-m)}}{r_m + \kappa(N-m)} \right]^2. \tag{77}$$

It follows by integrating the square root of each pair of exponential waiting densities. Writing $r_m = \kappa(N-m)e^{\xi_m}$ gives $F_{\exp} = \prod_m \operatorname{sech}^2(\xi_m/2)$. Thus small independent relative mismatches accumulate quadratically. This is a comparison-mode formula, not a necessary error budget for the fully optimized fidelity.

Proposition 12 (Exact-purity penalty under a native-rate cap)**.** *Assume the control can only attenuate native rates:*

$$0 \le r_m(t) \le \Gamma(m+1)(N-m). \tag{78}$$

If its lossless output is an exact single-mode Fock state, then $h(t) \le \Gamma$ and

$$\mathbb{E}T_{\mathrm{last}} \ge \frac{\mathsf{H}_N}{\Gamma}, \qquad \mathsf{H}_N = \sum_{j=1}^{N} \frac{1}{j}. \tag{79}$$

Constant $h = \Gamma$ attains this bound. Native emission has

$$\mathbb{E}T_{\mathrm{last}}^{\mathrm{native}} = \frac{2\mathsf{H}_N}{\Gamma(N+1)}. \tag{80}$$

If independent local loss γ is added to these same lossless-exact controls, their all-desired sector remains single-mode, but its probability is at most $(1+\gamma/\Gamma)^{-N}$.

Proof. The $m = 0$ constraint in (78), together with (74), gives $h \le \Gamma$. The zero-count state remains reachable whenever $S(t) > 0$, so it cannot be removed from this argument by appealing to typical counts. The cumulative hazard obeys $\int_0^t h \le \Gamma t$. Coupling each of the N independent target arrival times to a unit-rate exponential shows that it is no earlier than an exponential of

rate Γ. The expected maximum of the latter is H_N/Γ, proving (79). Summing the native waiting means $\{\Gamma j(N-j+1)\}^{-1}$ proves (80).

With local loss, the unnormalized one-photon mode is

$$v_\gamma(t) = e^{i\arg f(t)}\sqrt{h(t)}e^{-\frac{1}{2}\int_0^t h(s)\,\mathrm{d}s-\gamma t/2}, \qquad q = \|v_\gamma\|_2^2. \tag{81}$$

The all-desired field is $q^{N/2}|N_{v_\gamma/\sqrt{q}}\rangle$, with probability q^N. Since its lossless arrival time is stochastically no earlier than $\mathrm{Exp}(\Gamma)$, $q = \mathbb{E}e^{-\gamma t} \le \Gamma/(\Gamma+\gamma)$. Equality holds for constant $h = \Gamma$. □

The exact lossless-purity constraint therefore incurs at least a factor $(N+1)/2$ in mean completion time under the cap. This statement does not classify controls designed only to purify a lossy postselected sector, whose conditioning changes transition hazards. It also does not extend to approximate purity. The next construction shows why that distinction is essential.

12 A capped coherent control with asymptotically unit fidelity

Choose an even integer $4 \le K < N$ and set $T_0 = -\Gamma^{-1}\ln(1-K/N)$. Define a target cumulative distribution and its hazard by

$$u(t) = \begin{cases} 1-e^{-\Gamma t}, & 0 \le t \le T_0, \\ \left[1+\dfrac{N-K}{K}e^{-\Gamma N(t-T_0)/2}\right]^{-1}, & t \ge T_0, \end{cases} \tag{82}$$

$$h(t) = \frac{u'(t)}{1-u(t)} = \begin{cases} \Gamma, & t < T_0, \\ \Gamma N u(t)/2, & t > T_0. \end{cases} \tag{83}$$

The target mode is $f = \sqrt{u'}$. Its ideal Fock ladder would have rates $s_m = (N-m)h$. Instead implement the single coherent jump with zero phases and rates

$$r_m^C(t) = (N-m)\min\{h(t), \Gamma(m+1)\}. \tag{84}$$

They obey

$$\Gamma(N-m) \le r_m^C(t) \le \Gamma(m+1)(N-m). \tag{85}$$

Every rung is attenuated from its native value by the amplitude factor $\min\{1, \sqrt{h(t)/[\Gamma(m+1)]}\}$. This is coherent, excitation-dependent coupling to one output, not feedback based on a measured photon record. The piecewise regular model permits the hazard jump at T_0.

Theorem 13 (Finite-K fidelity and release-time guarantee)**.** *Let*

$$\delta_K = \min\left\{1, \frac{e^{-K/8}}{1-e^{-1/4}}\right\}. \tag{86}$$

The lossless source (84) *satisfies*

$$F_N^C[f] \ge (1-\delta_K)^2. \tag{87}$$

For $0 < \delta_{\mathrm{tail}} < 1$*, define*

$$T_1 = T_0 + \frac{2}{\Gamma N}\ln\frac{N(N-K)}{K\delta_{\mathrm{tail}}}. \tag{88}$$

Then

$$\mathbb{E}_C T_{\mathrm{last}} \le T_1 + (\delta_K + \delta_{\mathrm{tail}})\frac{\mathsf{H}_N}{\Gamma}. \tag{89}$$

For the smallest even $K \ge 24\ln N$ *and* $\delta_{\mathrm{tail}} = N^{-3}$*, at sufficiently large* N*,*

$$F_N^C[f] \ge 1 - O(N^{-3}), \qquad \mathbb{E}_C T_{\mathrm{last}} = O\left(\frac{\ln N}{\Gamma N}\right). \tag{90}$$

Proof. Under the target Fock density, $U_i = u(t_i)$ are independent uniform variables. Write $M(v) = \#\{i : U_i \le v\}$ and define

$$G = \{M(v) + 1 \ge Nv/2 \text{ for every } v \in [K/N, 1]\}. \tag{91}$$

On G, actual and target rates agree at every time along the entire history, including all no-jump intervals. Before T_0 they agree without restriction. Their positive amplitudes and probability densities therefore coincide pointwise on G.

If G fails at v, let $j = M(v) + 1 < Nv/2$. When $j \le K/2$, it follows that $M(K/N) < K/2$; the binomial lower-tail bound is $e^{-K/8}$. For $j > K/2$, the jth order statistic satisfies $U_{(j)} > 2j/N$, so a binomial with mean $2j$ is at most $j - 1$, of probability at most $e^{-j/4}$. Only $j < N/2$ can violate the inequality, but extending the sum to infinity increases the bound. Hence

$$\mathbb{P}_f(G^c) \le e^{-K/8} + \sum_{j>K/2} e^{-j/4} = \frac{e^{-K/8}}{1 - e^{-1/4}} \le \delta_K \tag{92}$$

when the displayed untruncated bound is at most one; otherwise the trivial probability bound supplies $\delta_K = 1$. The normalized actual and target densities agree on G, so $\mathbb{P}_C(G^c) = \mathbb{P}_f(G^c) \le \delta_K$. Their affinity is at least their common probability on G. Squaring proves (87).

Equation (88) gives $1 - u(T_1) = \delta_{\text{tail}}/(N + \delta_{\text{tail}})$. Thus the target probability of a late photon is $1 - u(T_1)^N \le N[1 - u(T_1)] \le \delta_{\text{tail}}$. Equality of the good-history densities implies $\mathbb{P}_C(T_{\text{last}} > T_1) \le \delta_K + \delta_{\text{tail}}$. From any state and time, the lower rate bound in (85) limits the mean remaining release time by H_N/Γ. Splitting the mean completion time at T_1 proves (89). Finally the stated choice of K gives $\delta_K = O(N^{-3})$, while $T_0 = O(\ln N/(\Gamma N))$ and the second term in (88) has the same order. □

The initial part of the target pulse contains mean count K. The later hazard rises with its deterministic cumulative probability. Only strongly delayed count histories would need a rate enhancement; clipping changes these rare histories. Thus the exact-purity slowdown in Section 11 is not a bound on fidelities tending to one. Equations (84)–(90) give a constructive counterexample under the very same native-rate cap.

12.1 Fixed local loss: polynomial acceptance and high conditional purity

Add independent local decay at rate γ per excitation. Let ψ_C be the normalized lossless capped-source amplitude. The all-desired amplitude is $e^{-\gamma \sum_i t_i/2}\psi_C$, by the same integrated-excitation identity used in Section 8. Define

$$q(\gamma) = \int_0^\infty e^{-\gamma t}|f(t)|^2\,\mathrm{d}t, \qquad g_\gamma(t) = q(\gamma)^{-1/2}e^{-\gamma t/2}f(t). \tag{93}$$

Theorem 14 (Loss robustness of the capped construction)**.** *Let P_{acc} be the probability that all N photons enter the desired output, and $Y_N[g_\gamma]$ its unconditional overlap with $|N_{g_\gamma}\rangle$. Then*

$$(1 - \delta_K)q(\gamma)^N \le P_{\text{acc}} \le q(\gamma)^N, \tag{94}$$

$$Y_N[g_\gamma] \ge q(\gamma)^N(1 - \delta_K)^2, \tag{95}$$

$$F_{N|\text{all desired}}[g_\gamma] \ge (1 - \delta_K)^2. \tag{96}$$

Moreover, with $\epsilon = \gamma/\Gamma$,

$$q(\gamma)^N \ge \exp\{-\epsilon[K + 2\ln(N/K)]\}. \tag{97}$$

For fixed ϵ and the $K = O(\ln N)$ choice above, acceptance has a polynomial lower bound in N, while conditional fidelity tends to one.

Proof. Exponential tilting by $e^{-\gamma t}$ makes the distribution $|g_\gamma|^2$ stochastically earlier than $|f|^2$. One may see this by noting that an increasing indicator of a late time has nonpositive covariance with the decreasing weight. The event G is preserved by moving any arrival earlier. An independent quantile coupling therefore gives $\mathbb{P}_{g_\gamma}(G) \geq 1 - \delta_K$.

On G, $\psi_C = \prod_i f(t_i)$. Integrating its locally attenuated squared modulus over G gives $P_{\mathrm{acc}} \geq q^N \mathbb{P}_{g_\gamma}(G)$. Likewise, restricting the positive overlap with $\prod_i g_\gamma(t_i)$ to G gives amplitude at least $q^{N/2}\mathbb{P}_{g_\gamma}(G)$, proving the yield bound.

For the upper bound on acceptance, $r_m^C(t) \leq s_m(t)$ at every common state and time. Couple the two pure-birth processes so that whenever their counts agree all actual jumps can be shared with target jumps, with extra jumps allowed for the target. When the counts differ, unit jumps cannot reverse their order. This gives every actual arrival time no earlier than its coupled target arrival. Hence

$$P_{\mathrm{acc}} = \mathbb{E}_C e^{-\gamma \sum_i t_i} \leq \mathbb{E}_f e^{-\gamma \sum_i t_i} = q^N. \tag{98}$$

Dividing the yield lower bound by this upper bound proves (96).

Integrating the survival function in (82) gives the exact first moment

$$N\mathbb{E}_f t = N \int_0^\infty [1 - u(t)]\, \mathrm{d}t = \frac{K + 2\ln(N/K)}{\Gamma}. \tag{99}$$

Jensen's inequality, $\mathbb{E}_f e^{-\gamma t} \geq e^{-\gamma \mathbb{E}_f t}$, proves (97). □

The polynomial exponent in (97) is a guarantee, not an optimized loss frontier. It may be worse than the native success exponent in (60). The result concerns the all-desired sector; operational use of its conditioning has the heralding qualification stated in Section 8.

Lemma 15 (Accepted completion times are stochastically earlier)**.** *For any such time-dependent pure-birth ladder with positive cumulative rates and complete release, conditioning on no independent local loss makes the completion time stochastically no later than its lossless completion time. In particular,*

$$\mathbb{E}[T_{\mathrm{last}} \mid \text{all desired}] \leq \mathbb{E}_C T_{\mathrm{last}}. \tag{100}$$

Proof. Let $R_m(t) = \int_0^t r_m(s)\, \mathrm{d}s$. Using independent unit-rate exponentials E_m, construct successive arrivals by

$$T_{m+1} = R_m^{-1}\big(R_m(T_m) + E_m\big), \qquad T_0 = 0. \tag{101}$$

Each T_j is coordinatewise increasing in all the E_m. Independent product measures are associated: two increasing functions have nonnegative covariance. This follows by the one-variable Chebyshev covariance identity and induction over independent coordinates. Therefore any increasing function of the arrivals has nonpositive covariance with $W = e^{-\gamma \sum_j T_j}$, a decreasing function of those coordinates. The no-local-loss conditional measure weights histories by $W/\mathbb{E}W$. Applying the covariance inequality to $\mathbf{1}_{\{T_N > t\}}$ proves stochastic domination; integrating the tail inequality proves (100). □

The argument uses monotonic cumulative rates and their inverses, not monotonic instantaneous rates in time. Consequently accepted runs of the capped construction also finish in mean time $O(\ln N/(\Gamma N))$.

13 Reproducible numerical consistency checks

The executable `verification/check_fidelity.py` evaluates specified trial modes, verifies exact identities, and reports ordinary floating-point convergence checks. Numerical overlaps are

achievable values for those modes, not certificates of global optimality. The analytic uniform and asymptotic theorems do not depend on fitting the numerical results.

For these numerical calculations set $\Gamma = 1$. For the native source, use scaled time $x = Nt$ and write $f(t) = \sqrt{N}\,\phi(x)$. Set $c_0(0) = 1$ and $c_j(0) = 0$ for $j > 0$. The positive ordered-amplitude recursion is

$$c_j' = -\frac{(j+1)(N-j)}{2N} c_j + j\sqrt{\frac{N-j+1}{N}}\,\phi(x) c_{j-1}, \qquad j = 0, \ldots, N, \tag{102}$$

with the second term absent at $j = 0$. Then $c_N(\infty)$ is the exact trial amplitude. Thus an ordered N-dimensional integral reduces to $N + 1$ ordinary differential equations. The same construction with the general cascade rates evaluates the partial-source and local-loss conditional kernels. The reported values use the trial mode truncated at $x = X = \ln(M/a) + 35$ and then exactly renormalized. If q is the removed one-photon norm, then

$$q = \frac{M e^{-X}}{a + (n-1)e^{-X}}. \tag{103}$$

The infinite and truncated n-photon target states have squared overlap $(1-q)^n$. Their projection probabilities against any source therefore differ by at most $\sqrt{1-(1-q)^n}$, their pure-state trace distance. The code reports this conservative analytic cutoff budget separately from the measured change between two numerical cutoffs.

For the explicit fully inverted pulse (11), representative values are

N	Trial fidelity	$\sqrt{N}$ times trial fidelity	Infinite-pulse lower bound
1	1.000000000	1.000000	1.000000000
2	0.955818490	1.351731	0.923632012
5	0.833841887	1.864527	0.762238935
10	0.704689065	2.228422	0.618866167
20	0.565254318	2.527894	0.480738698
50	0.396241902	2.801853	0.327835432
100	0.292837776	2.928378	0.239469289

The infinite one-photon pulse has fidelity one exactly. The exact two-photon optimum in Section 6 is larger than the trial value, as required.

For the partial-source trial pulse (30), the check also evaluates the relative entropy by an independent one-dimensional quantile integral:

M	n	$e^{-D_{M,n}}$	Trial fidelity
10	2	0.998151912	0.999058744
20	10	0.967523540	0.981861157
100	10	0.999176823	0.999581755
100	90	0.681685371	0.763737854

A noninteger-parameter example is native emission with $N = 5$, $\epsilon = 0.25$. Equation (56) gives $p_0 = 0.5883789413$; the conditional trial fidelity is 0.8693225961, and the unconditional trial yield is 0.5114911087.

Across these default rows, the largest change between the two tolerance settings is 2.7×10^{-12}, and the largest change between tail parameters 30 and 35 is 6.5×10^{-12}. The largest mode-normalization residual is 2.2×10^{-16}, and the independent quantile-integral discrepancy is

5.7×10^{-14}. Evaluating the exact two-photon Bessel mode with the cascade recursion gives a squared-fidelity residual of 1.9×10^{-13}.

The verification also compares the direct product and gamma quotient for p_0, the unweighted hypergeometric identity $\sum_m \mathrm{HG}(N, m, k, j) = (N+1)/(k+1)$, and the weighted identity used in (52). The hypergeometric comparisons use exact rational arithmetic. The program's machine-readable output records the residuals, tolerance comparisons, and cutoff budgets. These checks support implementation consistency; the analytic proofs establish the uniform and asymptotic results.

14 Relation to prior work and physical scope

The collective rate ladder, multiphoton wavefunction, and multimode character of the emitted field are established background. Paulisch and collaborators analyzed superradiant photonic states and their metrological usefulness [1]. Perarnau-Llobet and collaborators developed efficient observables and explicit few-mode Fock decompositions [2]. Lemberger and Mølmer studied the one-photon temporal modes and found a dominant occupation above 90 percent [3]. That occupation fraction does not determine the optimized probability that all photons share one mode.

Tziperman and collaborators studied complete propagating quantum states and finite-size single-mode Fock fidelities, including direct collective decay [5]. Liu and Shen's weak-excitation Fock-generation regime [6] is consistent with the dilute symmetric-source bounds above. The discrete cavity-mode Tavis–Cummings setting of Ref. [8] is a different output problem. Belliardo and collaborators optimize temporal modes for information extraction from a superradiant burst [9]; an information-optimal mode can contain a vanishing fraction of the photons. A task-optimal mode therefore need not be the most occupied one-photon mode.

Quantum-soliton analyses describe how center-of-mass delocalization can suppress a many-body Hartree overlap even when its Nth root stays close to one [7]. That root is distinct from a one-photon occupation or reduced-state fidelity. The pair-distance factor in Eq. (7) makes this comparison relevant. Malz, Trivedi, and Cirac established the large-N Dicke population dynamics [4]; the count argument above starts from that collective rate model and proves a uniform arrival-density bound.

Deterministic optical Fock preparation, shaped bosonic input synthesis, and controlled cavity release are established constructions [11, 10, 12]. Dissipative Rydberg filtering, photon subtraction, and temporal heralding provide other state-preparation mechanisms [14, 13, 15]. They motivate distinctions among deterministic emission, conditional fidelity, and unconditional target yield.

Transition-rate engineering along collective ladders is used, for example, in the superabsorption proposal of Higgins and collaborators [16]. Resolving transition frequencies and engineering their rates does not alone implement the coherent single jump $\sum_m \sqrt{r_m(t)}|m+1\rangle\langle m|$ required by the capped construction. Distinct output channels can retain rung information. The construction specifies a coherent excitation-dependent coupling to the same output, with no additional environmental history record. It is not an experimental architecture or a finite-bandwidth implementation.

The proved restrictions depend on the allowed operations. Uniform attenuation, symmetric diagonal phases, and deterministic common rate shaping retain the modal ceiling in their stated senses. Prefix detection changes the photon-number resource. Excitation-dependent coherent coupling changes the transition ladder. Exact Fock factorization under a native-rate cap is slow, while the controlled approximation in Section 12 retains fast release and fidelity tending to one. These results do not classify all excitation-preserving dynamics, other spin sectors, ancillary

memories, adaptive measurements, or non-Markov sources. Emitter disorder, additional channels, and unobserved local-loss histories require their own output-state analysis.

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